

Assignment No 3

Employee Performance Appraisal System

1) Single - Row Subquery

```
mysql> SELECT Emp_Name, Salary
-> FROM Employee
-> WHERE Salary = (SELECT MAX(Salary) FROM Employee);
+-----+-----+
| Emp_Name | Salary |
+-----+-----+
| Vijay    | 50000.00 |
| Prachi   | 50000.00 |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT Emp_Name, Salary
-> FROM Employee
-> WHERE Salary > (SELECT AVG(Salary) FROM Employee);
+-----+-----+
| Emp_Name | Salary |
+-----+-----+
| Vijay    | 50000.00 |
| Prachi   | 50000.00 |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT Emp_Name, Salary
-> FROM Employee
-> WHERE Salary < (SELECT AVG(Salary) FROM Employee);
+-----+-----+
| Emp_Name | Salary |
+-----+-----+
| Atharva  | 30000.00 |
| Omraje   | 30000.00 |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT Emp_Name
-> FROM Employee
-> WHERE Salary >=
-> (SELECT AVG(Salary) FROM Employee);
+-----+
| Emp_Name |
+-----+
| Vijay    |
| Sahil    |
| Prachi   |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT Emp_Name
-> FROM Employee
-> WHERE Salary <=
-> (SELECT MAX(Salary) FROM Employee);
+-----+
| Emp_Name |
+-----+
| Vijay    |
| Sahil    |
| Atharva  |
| Omraje   |
| Prachi   |
+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT Emp_Name
-> FROM Employee
-> WHERE Salary <>
-> (SELECT MAX(Salary) FROM Employee);
+-----+
| Emp_Name |
+-----+
| Sahil    |
| Atharva  |
| Omraje   |
+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT Emp_Name
-> FROM Employee
-> WHERE Salary !=
-> (SELECT MIN(Salary) FROM Employee);
```

Emp_Name
Vijay
Sahil
Prachi

```
3 rows in set (0.00 sec)
```

2) Multiple - Row Subquery

```
mysql> SELECT Emp_Name
-> FROM Employee
-> WHERE Dept_ID IN
-> (SELECT Dept_ID
-> FROM Dept
-> WHERE Location = 'Head Office');
```

Emp_Name
Prachi

```
1 row in set (0.00 sec)
```

```
mysql> SELECT Emp_Name
-> FROM Employee
-> WHERE Dept_ID NOT IN
-> (SELECT Dept_ID
-> FROM Dept
-> WHERE Location = 'Head Office');
```

Emp_Name
Vijay
Sahil
Atharva
Omraje

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT Emp_Name, Salary
-> FROM Employee
-> WHERE Salary > Any
-> ( SELECT Salary
-> FROM Employee
-> WHERE Dept_Id = 2);
```

Emp_Name	Salary
Vijay	50000.00
Prachi	50000.00

2 rows in set (0.00 sec)

```
mysql> SELECT Emp_Name, Salary
-> FROM Employee
-> WHERE Salary > SOME
-> ( SELECT Salary
-> FROM Employee
-> WHERE Dept_ID = 3);
```

Emp_Name	Salary
Vijay	50000.00
Sahil	40000.00
Prachi	50000.00

3 rows in set (0.00 sec)

```
mysql> SELECT Emp_Name, Salary
-> FROM Employee
-> WHERE Salary > ALL
-> ( SELECT Salary
-> FROM Employee
-> WHERE Dept_ID = 3);
```

Emp_Name	Salary
Vijay	50000.00
Sahil	40000.00
Prachi	50000.00

3 rows in set (0.00 sec)

3) Exists Operator

```
mysql> SELECT Emp_Name
-> FROM Employee e
-> WHERE EXISTS
-> (SELECT 1
-> FROM Appraisal a
-> WHERE a.Emp_ID = e.Emp_ID);
+-----+
| Emp_Name |
+-----+
| Vijay    |
| Sahil    |
| Atharva  |
+-----+
3 rows in set (0.00 sec)

mysql> SELECT Emp_Name
-> FROM Employee e
-> WHERE NOT EXISTS
-> (SELECT 1
-> FROM Appraisal a
-> WHERE a.Emp_ID = e.Emp_ID);
+-----+
| Emp_Name |
+-----+
| Omraje   |
| Prachi   |
+-----+
2 rows in set (0.00 sec)
```

4) Multiple Column SubQuery

```
mysql> SELECT Emp_Name
-> FROM Employee
-> WHERE (Dept_ID, Salary) IN
-> (
->     SELECT e.Dept_ID, e.Salary
->     FROM Employee e
->     JOIN Appraisal a
->     ON e.Emp_ID = a.Emp_ID
->     WHERE a.rating = 'Outstanding'
-> );
+-----+
| Emp_Name |
+-----+
| Vijay    |
+-----+
1 row in set (0.02 sec)
```

5) Correlated SubQuery

```
mysql> SELECT Emp_Name, Salary, Dept_ID
-> FROM Employee e1
-> WHERE Salary >
-> (
->     SELECT AVG(Salary)
->     FROM Employee e2
->     WHERE e2.Dept_ID = e1.Dept_ID
-> );
```

Emp_Name	Salary	Dept_ID
Rohan	60000.00	1

1 row in set (0.00 sec)

6) Nested SubQuery

```
mysql> SELECT Emp_Name
-> FROM Employee
-> WHERE Dept_ID IN
-> (
->     SELECT Dept_ID
->     FROM Dept
->     WHERE Location =
->     (
->         SELECT Location
->         FROM Dept
->         WHERE Dept_name = 'IT'
->     )
-> );
```

Emp_Name
Sahil

1 row in set (0.00 sec)

7) Joins

i) Inner Join

```
mysql> SELECT e.Emp_Name, d.Dept_name, d.Location
-> FROM Employee e
-> INNER JOIN Dept d
-> ON e.Dept_ID = d.Dept_ID;
```

Emp_Name	Dept_name	Location
Vijay	HR	Mumbai
Sahil	IT	Pune
Atharva	Finance	Delhi
Omraje	AIML	Sangli
Prachi	Developer	Head Office
Rohan	HR	Mumbai

6 rows in set (0.00 sec)

ii) Left Join

```
mysql> SELECT e.Emp_Name, d.Dept_name
-> FROM Employee e
-> LEFT JOIN Dept d
-> ON e.Dept_ID = d.Dept_ID;
```

Emp_Name	Dept_name
Vijay	HR
Sahil	IT
Atharva	Finance
Omraje	AIML
Prachi	Developer
Rohan	HR

6 rows in set (0.00 sec)

iii) Right Join

```
mysql> SELECT e.Emp_Name, d.Dept_name
-> FROM Employee e
-> RIGHT JOIN Dept d
-> ON e.Dept_ID = d.Dept_ID;
```

Emp_Name	Dept_name
Vijay	HR
Rohan	HR
Sahil	IT
Atharva	Finance
Omraje	AIML
Prachi	Developer
NULL	NULL
NULL	NULL
NULL	SALES

9 rows in set (0.00 sec)

iv) Full Join

```
mysql> SELECT e.Emp_Name, d.Dept_name
-> FROM Employee e
-> LEFT JOIN Dept d
-> ON e.Dept_ID = d.Dept_ID
->
-> UNION
->
-> SELECT e.Emp_Name, d.Dept_name
-> FROM Employee e
-> RIGHT JOIN Dept d
-> ON e.Dept_ID = d.Dept_ID;
```

Emp_Name	Dept_name
Vijay	HR
Sahil	IT
Atharva	Finance
Omraje	AIML
Prachi	Developer
Rohan	HR
NULL	NULL
NULL	NULL
NULL	SALES

9 rows in set (0.01 sec)

v) Cross Join

```
mysql> SELECT e.Emp_Name, d.Dept_name  
-> FROM Employee e  
-> CROSS JOIN Dept d;
```

Emp_Name	Dept_name
Rohan	HR
Prachi	HR
Omraje	HR
Atharva	HR
Sahil	HR
Vijay	HR
Rohan	IT
Prachi	IT
Omraje	IT
Atharva	IT
Sahil	IT
Vijay	IT
Rohan	Finance
Prachi	Finance
Omraje	Finance
Atharva	Finance
Sahil	Finance
Vijay	Finance
Rohan	AIML
Prachi	AIML
Omraje	AIML
Atharva	AIML
Sahil	AIML
Vijay	AIML
Rohan	Developer
Prachi	Developer
Omraje	Developer
Atharva	Developer
Sahil	Developer
Vijay	Developer
Rohan	NULL
Prachi	NULL
Omraje	NULL
Atharva	NULL
Sahil	NULL
Vijay	NULL

Rohan	
Prachi	
Omraje	
Atharva	
Sahil	
Vijay	
Rohan	SALES
Prachi	SALES
Omraje	SALES
Atharva	SALES
Sahil	SALES
Vijay	SALES

48 rows in set (0.01 sec)

vi) Self Join

```
mysql> SELECT e1.Emp_Name AS Employee1,
-> e2.Emp_Name AS Employee2,
-> e1.Dept_ID
-> FROM Employee e1
-> JOIN Employee e2
-> ON e1.Dept_ID = e2.Dept_ID
-> AND e1.Emp_ID <> e2.Emp_ID;
```

Employee1	Employee2	Dept_ID
Vijay	Rohan	1
Rohan	Vijay	1

2 rows in set (0.00 sec)

8) Comparison Operators

```
mysql> SELECT * FROM Employee
-> WHERE Salary = 50000;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
105	Prachi	Manager	50000.00	5

2 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Dept_ID <> 1;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
102	Sahil	Developer	40000.00	2
103	Atharva	Analyst	30000.00	3
104	Omraje	Developer	30000.00	4
105	Prachi	Manager	50000.00	5

4 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Salary != 30000;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
102	Sahil	Developer	40000.00	2
105	Prachi	Manager	50000.00	5
106	Rohan	HR Exec	60000.00	1

4 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Salary > 40000;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
105	Prachi	Manager	50000.00	5
106	Rohan	HR Exec	60000.00	1

3 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Salary >= 40000;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
102	Sahil	Developer	40000.00	2
105	Prachi	Manager	50000.00	5
106	Rohan	HR Exec	60000.00	1

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Employee
-> WHERE Salary <= 30000;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
103	Atharva	Analyst	30000.00	3
104	Omraje	Developer	30000.00	4

```
2 rows in set (0.00 sec)
```

9) Logical Operators

```
mysql> SELECT * FROM Employee
-> WHERE Salary >= 40000 AND Dept_ID = 1;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
106	Rohan	HR Exec	60000.00	1

```
2 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Employee
-> WHERE Dept_ID = 1 OR Dept_ID = 2;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
106	Rohan	HR Exec	60000.00	1
102	Sahil	Developer	40000.00	2

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Employee
-> WHERE NOT Salary = 30000;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
102	Sahil	Developer	40000.00	2
105	Prachi	Manager	50000.00	5
106	Rohan	HR Exec	60000.00	1

4 rows in set (0.00 sec)

10) Range Operators

```
mysql> SELECT * FROM Employee
-> WHERE Salary BETWEEN 30000 AND 50000;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
102	Sahil	Developer	40000.00	2
103	Atharva	Analyst	30000.00	3
104	Omraje	Developer	30000.00	4
105	Prachi	Manager	50000.00	5

5 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Salary NOT BETWEEN 30000 AND 40000;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
105	Prachi	Manager	50000.00	5
106	Rohan	HR Exec	60000.00	1

3 rows in set (0.00 sec)

11) Set Operators

```
mysql> SELECT * FROM Employee
-> WHERE Dept_ID IN (1,2,3);
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
106	Rohan	HR Exec	60000.00	1
102	Sahil	Developer	40000.00	2
103	Atharva	Analyst	30000.00	3

4 rows in set (0.01 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Dept_ID NOT IN (1,2,3);
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
104	Omraje	Developer	30000.00	4
105	Prachi	Manager	50000.00	5

2 rows in set (0.00 sec)

12) Pattern Matching Operators

```
mysql> SELECT * FROM Employee
-> WHERE Emp_Name LIKE 'A%';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
103	Atharva	Analyst	30000.00	3

1 row in set (0.01 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Emp_Name LIKE '%a';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
103	Atharva	Analyst	30000.00	3

1 row in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Emp_Name LIKE '%a%';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
102	Sahil	Developer	40000.00	2
103	Atharva	Analyst	30000.00	3
104	Omraje	Developer	30000.00	4
105	Prachi	Manager	50000.00	5
106	Rohan	HR Exec	60000.00	1

6 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Emp_Name LIKE '_a%';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
102	Sahil	Developer	40000.00	2

1 row in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Emp_Name LIKE '____';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
104	Omraje	Developer	30000.00	4
105	Prachi	Manager	50000.00	5

2 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Emp_Name NOT LIKE 'A%';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
102	Sahil	Developer	40000.00	2
104	Omraje	Developer	30000.00	4
105	Prachi	Manager	50000.00	5
106	Rohan	HR Exec	60000.00	1

5 rows in set (0.00 sec)

13) Null Operators

```
mysql> SELECT * FROM Dept
-> WHERE Dept_name IS NULL;
```

Dept_ID	Dept_name	Location
6	NULL	Head Office

1 row in set (0.00 sec)

```
mysql> SELECT * FROM Dept
-> WHERE Dept_name IS NOT NULL;
```

Dept_ID	Dept_name	Location
1	HR	Mumbai
2	IT	Pune
3	Finance	Delhi
4	AIML	Sangli
5	Developer	Head Office
7		Head Office
8	SALES	Head Office

7 rows in set (0.00 sec)

14) Subquery Operators

```
mysql> SELECT * FROM Employee e
-> WHERE EXISTS
-> (
->   SELECT 1 FROM Appraisal a
->   WHERE a.Emp_ID = e.Emp_ID
-> );
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
102	Sahil	Developer	40000.00	2
103	Atharva	Analyst	30000.00	3

3 rows in set (0.01 sec)

```
mysql> SELECT * FROM Employee e
-> WHERE NOT EXISTS
-> (
->     SELECT 1 FROM Appraisal a
->     WHERE a.Emp_ID = e.Emp_ID
-> );
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
104	Omraje	Developer	30000.00	4
105	Prachi	Manager	50000.00	5
106	Rohan	HR Exec	60000.00	1

3 rows in set (0.01 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Salary > ANY
-> (
->     SELECT Salary FROM Employee WHERE Dept_ID = 2
-> );
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
105	Prachi	Manager	50000.00	5
106	Rohan	HR Exec	60000.00	1

3 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Salary > SOME
-> (
->     SELECT Salary FROM Employee WHERE Dept_ID = 3
-> );
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
102	Sahil	Developer	40000.00	2
105	Prachi	Manager	50000.00	5
106	Rohan	HR Exec	60000.00	1

4 rows in set (0.00 sec)

15) Boolean Operators

```
mysql> SELECT * FROM Appraisal
-> WHERE performance_score IS TRUE;
```

Apr_ID	Emp_ID	Apr_year	performance_score	rating
1	101	2024	9	Outstanding
2	102	2024	8	Very Good
3	103	2024	6	Good

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Employee
-> WHERE (Salary > 40000) IS FALSE;
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
102	Sahil	Developer	40000.00	2
103	Atharva	Analyst	30000.00	3
104	Omraje	Developer	30000.00	4

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Dept
-> WHERE (Dept_Name = 'HR') IS UNKNOWN;
```

Dept_ID	Dept_name	Location
6	NULL	Head Office

```
1 row in set (0.00 sec)
```

16) Regular Expressions

```
mysql> SELECT * FROM Employee
-> WHERE Emp_Name REGEXP 'ra';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
104	Omraje	Developer	30000.00	4
105	Prachi	Manager	50000.00	5

2 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Emp_Name REGEXP '^[SA]';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
102	Sahil	Developer	40000.00	2
103	Atharva	Analyst	30000.00	3

2 rows in set (0.02 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Emp_Name REGEXP '^.....$';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
101	Vijay	SALES	50000.00	1
102	Sahil	Developer	40000.00	2
106	Rohan	HR Exec	60000.00	1

3 rows in set (0.00 sec)

```
mysql> SELECT * FROM Employee
-> WHERE Emp_Name LIKE 'a%';
```

Emp_ID	Emp_Name	Designation	Salary	Dept_ID
103	Atharva	Analyst	30000.00	3

1 row in set (0.00 sec)

17) Arithmetic Operators

```
mysql> SELECT Emp_Name, Salary
-> FROM Employee
-> WHERE Salary + 5000 > 45000;
+-----+-----+
| Emp_Name | Salary |
+-----+-----+
| Vijay    | 50000.00 |
| Prachi   | 50000.00 |
| Rohan    | 60000.00 |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT Emp_Name, Salary
-> FROM Employee
-> WHERE Salary - 5000 < 35000;
+-----+-----+
| Emp_Name | Salary |
+-----+-----+
| Atharva  | 30000.00 |
| Omraje   | 30000.00 |
+-----+-----+
2 rows in set (0.00 sec)
```

```
mysql> SELECT Emp_Name, Salary
-> FROM Employee
-> WHERE Salary * 12 > 500000;
+-----+-----+
| Emp_Name | Salary |
+-----+-----+
| Vijay    | 50000.00 |
| Prachi   | 50000.00 |
| Rohan    | 60000.00 |
+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> SELECT Emp_Name, Salary  
-> FROM Employee  
-> WHERE Salary / 2 > 20000;
```

```
+-----+-----+  
| Emp_Name | Salary |  
+-----+-----+  
| Vijay    | 50000.00 |  
| Prachi   | 50000.00 |  
| Rohan    | 60000.00 |  
+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> SELECT *  
-> FROM Employee  
-> WHERE Emp_ID % 2 = 0;
```

```
+-----+-----+-----+-----+-----+  
| Emp_ID | Emp_Name | Designation | Salary | Dept_ID |  
+-----+-----+-----+-----+-----+  
| 102    | Sahil    | Developer   | 40000.00 | 2 |  
| 104    | Omraje   | Developer   | 30000.00 | 4 |  
| 106    | Rohan    | HR Exec     | 60000.00 | 1 |  
+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> SELECT *  
-> FROM Employee  
-> WHERE Emp_ID % 2 != 0;
```

```
+-----+-----+-----+-----+-----+  
| Emp_ID | Emp_Name | Designation | Salary | Dept_ID |  
+-----+-----+-----+-----+-----+  
| 101    | Vijay    | SALES       | 50000.00 | 1 |  
| 103    | Atharva  | Analyst     | 30000.00 | 3 |  
| 105    | Prachi   | Manager     | 50000.00 | 5 |  
+-----+-----+-----+-----+-----+  
3 rows in set (0.00 sec)
```